

1 Snowmelt regime and dispersal limitation predict the
2 decadal expansion of dwarf bamboo into alpine snowbed
3 vegetation

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7 Abstract

8 Dwarf bamboo (*Sasa* spp.) is expanding into Japanese alpine snowbed vegeta-
9 tion, excluding subordinate species and threatening plant diversity. Mapping
10 its distributional change requires approximately 1 m resolution, because alpine
11 vegetation forms a fine mosaic along microtopographic gradients. Using 10 years
12 of ground-based time-lapse imagery of an alpine slope on Mt. Tateyama, Japan,
13 we produced 1 m vegetation maps for 2012 and 2021 and per-pixel snowmelt
14 dates for 2011–2021. Over 9 years, *Sasa* showed a net change of +19% (gross
15 expansion 48%, gross loss 29%, nearly half of it exchange with dwarf pine).
16 Because *Sasa* expands through both rhizome elongation and seed dispersal, we
17 fitted two models under a common spatial-block design: a habitat suitability
18 model of where conditions permit *Sasa* (Model A) and an establishment model
19 of the observed 2012→2021 transitions, including distance from the existing
20 front (Model B). In both models, snowmelt climatology (the 2011–2021 mean
21 snowmelt date) was among the strongest environmental predictors and indicated
22 a favourable snowmelt window (day of year approximately 150–185). Estab-
23 lishment probability over 9 years, however, was most strongly associated with
24 proximity to existing patches: approximately 10% within 5 m of the front and
25 below 1% beyond 10 m. Environmentally suitable area exceeded occupied area
26 approximately 16-fold, suggesting that dispersal constrains the current distri-
27 bution more than environment. A stochastic cellular automaton using Model
28 B reproduced the observed expansion and projected approximately 4,900 m²
29 of new establishment by 2030 under unchanged snowmelt and 5,700 m² under
30 continuation of the observed, statistically undetectable trend; these are scenarios,
31 not forecasts. Nine tenths of expected establishment within herbaceous vege-
32 tation fell in two snowbed communities adjoining existing patches. Near-term
33 expansion is therefore spatially predictable, making removal or containment at
34 fronts bordering snowbeds a high-priority candidate intervention, complemented
35 by surveillance of environmentally suitable areas for establishment from seed.

36 1. Introduction

37 Alpine ecosystems are among the most vulnerable to climate change (Hock
38 et al., 2019). Resurveys across mountain regions have documented upslope range
39 shifts, increasing summit species richness, and changes in the composition of
40 high-elevation vegetation (Gottfried et al., 2012; Rumpf et al., 2018; Lamprecht
41 et al., 2018), while satellite observations show widespread greening of alpine
42 landscapes (Carlson et al., 2017). Much of this change reflects the expansion of
43 species that gain a competitive advantage as environmental constraints relax.
44 Examples include woody shrub expansion in Arctic and alpine tundra (Myers-
45 Smith et al., 2011; Bjorkman et al., 2018) and the encroachment of grasses
46 and tall forbs into snowbed communities as the snow-free season lengthens
47 (Grabherr, 2003; Dullinger et al., 2003). In Japanese alpine ecosystems, dwarf
48 bamboo (*Sasa* spp., hereafter *Sasa*) represents a prominent example. *Sasa* forms
49 dense evergreen stands that exclude subordinate species, and its expansion into

snowbed meadows has been associated with earlier snowmelt and reductions in plant species richness (Kudo et al., 2011). Snowbed vegetation has recovered following experimental removal of *Sasa* (Kudo et al., 2017). Understanding where and how rapidly *Sasa* expands is therefore directly relevant to alpine plant conservation.

Snowmelt timing is a central environmental factor for alpine vegetation because it determines the onset and duration of the growing season and is closely linked to growing conditions such as soil temperature and moisture (Wipf and Rixen, 2010; Wahren et al., 2005); snowmelt date is therefore a proxy for a set of covarying conditions rather than a single driver. Spatial scale matters here. Snowmelt varies with microtopography over distances of a few to tens of metres, and alpine vegetation forms a fine mosaic of communities along these snowmelt gradients. Linking changes in alpine plant distributions to environmental conditions therefore requires simultaneous observations of vegetation and snowmelt at approximately 1 m resolution. Field plots capture this scale but lack continuous spatial coverage. Satellites provide decades of snow and vegetation observations at 10–30 m resolution, but cannot resolve community mosaics or movements of patch boundaries. Ground-based time-lapse cameras fill this gap. Repeated daily photographs from a fixed viewpoint allow automated extraction of both per-pixel snowmelt dates and vegetation maps at sub-metre ground resolution (Ide and Oguma, 2013; Okamoto et al., 2024). The two approaches are therefore complementary: the expansion patterns that satellites record over decades and at scales of tens of metres can be examined through observations of the metre-scale processes that underlie them.

Sasa expands through two contrasting dispersal processes (Suyama et al., 2000; Kudo et al., 2011). The first is the gradual advance of patch fronts through rhizome elongation; where this process dominates, expansion may be limited by the rate of dispersal rather than by the extent of environmentally suitable habitat. The second is establishment from seed following mass or partial flowering (Miyazaki et al., 2009; Mizuki et al., 2014), which can bypass local distance constraints and form new patches. Conservation planning requires information on both: where environmental conditions permit *Sasa* to establish if seeds arrive, and how far existing fronts can actually reach. For the former question, habitat suitability models, statistical frameworks relating species presence to environmental conditions, have been widely used in conservation planning and projections of distributional change (Guisan et al., 1998; Larson et al., 2004; Thomas et al., 2004; Gallien et al., 2010; Amagai et al., 2022). Projecting suitability maps without accounting for dispersal constraints, however, can greatly overestimate establishment area (Engler and Guisan, 2009; Engler et al., 2012). Answering the latter question requires distribution maps of the same area at two or more times. Comparing two maps reveals the cells in which establishment actually occurred. These observed transitions can then be modelled directly as a function of environment and distance from the existing front, projecting expansion as a process rather than as a reassessment of a static suitability map.

Here, we apply this approach to *Sasa* on the western slope of Mt. Tateyama in the northern Japanese Alps. Plot resurveys on the Murodo-daira plateau have documented 44–260% expansion of *Sasa* patches over 38 years (Yoshida et al., 2016). Using 10 years of time-lapse imagery, we (i) map *Sasa* distributions in 2012 and 2021 at 1 m resolution and quantify gross expansion, gross loss, and net change; (ii) characterise the snowmelt regime and its decadal trend for every pixel; (iii) fit a habitat suitability model of environmental potential (Model A) and an establishment model of observed transitions (Model B) to estimate how strongly snowmelt regime and topography are associated with *Sasa* distribution and establishment, while accounting for proximity to existing patches; (iv) project establishment to 2030 under explicit snowmelt scenarios using stochastic simulations based on the establishment model; and (v) identify the alpine communities most exposed to near-term expansion.

2. Materials and Methods

Figure 1 presents an overview of the workflow. Using time-lapse imagery, we (i) produced vegetation classification maps for 2012 and 2021 to quantify 9-year changes in *Sasa* distribution and (ii) characterised the snowmelt regime and its temporal trend from per-pixel snowmelt dates for 2011–2021, expressed as day of year (DOY). We then (iii) fitted two statistical models, a habitat suitability model of environmental potential (Model A) and an establishment model of the observed 2012–2021 transitions (Model B), and (iv) projected establishment to 2030 under explicit snowmelt scenarios using a stochastic cellular automaton based on Model B. All spatial analyses used a common 1 m grid (JGD2011 / UTM zone 53N, EPSG:6690).

2.1. Study site

The study slope lies on the western side of Mt. Tateyama (Murodo-daira and the surrounding slopes, 2,350–3,015 m a.s.l.) in the northern Japanese Alps, within Chubu-Sangaku National Park, Toyama Prefecture. The region is among the snowiest alpine environments in the world. Airborne laser surveys at peak snow accumulation show that mean snow depth on the western slope of Mt. Tateyama increases linearly with elevation, reaching approximately 6–7 m at Murodo-daira, with substantially greater local depths in snowdrifts (Iida et al., 2006). The plateau consists of pyroclastic deposits from Tateyama Volcano, and the slopes are dissected by avalanche tracks and hollows that retain snow. Snowmelt progresses from wind-exposed ridges, typically in May–June, to snowbed hollows where snow persists into August, creating steep gradients in growing-season length over horizontal distances of tens of metres.

The vegetation forms a mosaic typical of the snowy alpine zone of central Japan: dwarf pine scrub on exposed sites, snowbed communities in late-melting hollows, tall-forb meadows on moist slopes, and deciduous shrubs (rowans, maple, and montane alder) along streams and avalanche tracks. *Sasa* forms dense stands near the lower limit of the alpine zone and occurs as scattered patches within the

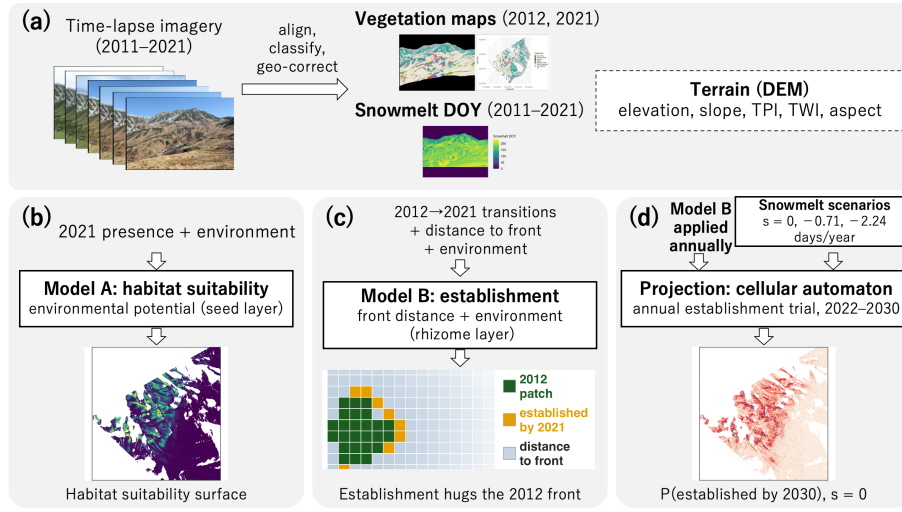


Figure 1: Images from a single fixed camera support two models corresponding to the two modes of *Sasa* expansion and projections of future establishment. (a) Data generation: alignment, classification, and georectification of time-lapse imagery produce vegetation maps (2012, 2021) and per-pixel snowmelt DOY (2011–2021). Terrain variables are derived from a DEM. (b) Model A: presence in 2021 and environmental conditions are used to estimate a habitat suitability surface (environmental potential; seed-dispersal stratum). (c) Model B: observed 2012→2021 transitions are modelled as a function of distance from the front and environment (rhizome stratum). Establishment is concentrated along the edges of 2012 patches, as illustrated by the orange cells in the schematic. (d) Projection: a stochastic cellular automaton applying Model B annually, updating the distance surface each year and taking snowmelt scenario s as input, yields establishment probabilities by 2030.

study slope. Although the area receives intensive tourist use via the Tateyama Kurobe Alpine Route, visitors remain on maintained trails, and the observed slope is not directly affected by trampling. No vegetation management or *Sasa* removal took place on the slope during the study period.

2.2. Image acquisition

We used images captured by a digital time-lapse camera (EOS 5D Mark II, Canon Inc., 21 megapixels) installed by the National Institute for Environmental Studies on the Murodo-sanso mountain lodge (approximately 2,450 m a.s.l.). Since 2010, the camera has photographed the slope hourly from 6:00 to 19:00 between April and November. For details of the installation and operation, see [Okamoto et al. \(2024\)](#). Vegetation classification used images acquired in August–October of 2012 and 2021 (7 frames per year), and snowmelt detection used images from April–August of each year during 2011–2021. Data for 2019 are missing because of construction work at the mountain lodge housing the camera. Images from 2010 were excluded because a different lens gave an incompatible field of view. Snowmelt detection followed the workflow of [Ide and Oguma \(2013\)](#), which is based on Otsu’s binarisation method ([Otsu, 1979](#)). Detectable snowmelt dates were censored by the imaging season (approximately DOY 120–230), and pixels without valid detections, including sky and permanently occluded terrain, were treated as missing.

2.3. Vegetation classification maps

Vegetation classification maps were produced using the three-step pipeline of [Okamoto et al. \(2024\)](#).

- **Image alignment.** All frames from both years were aligned to a common reference frame by matching AKAZE local features and estimating projective transformations with RANSAC in OpenCV. Median residual misalignment between the 2012 and 2021 series was 1–2 pixels, with no systematic offset between years.
- **Vegetation classification.** Each pixel was assigned to one of 7 classes: dwarf pine (*Pinus pumila*), dwarf bamboo (*Sasa* spp.), rowans (*Sorbus sambucifolia*, *S. matsumurana*), maple (*Acer tschonoskii*), montane alder (*Alnus viridis* subsp. *maximowiczii*), other vegetation (alpine herbs and dwarf shrubs such as crowberry and avens), and no vegetation. The classifier was a convolutional recurrent neural network (Conv2D–LSTM on 5×5-pixel patches) using temporal changes in leaf colour across seasonal image sequences. Training labels comprised 216 hand-drawn polygons, delineated mainly on 2015 imagery and checked against the 2012 and 2021 image sequences. Overall accuracy in 5-fold cross-validation was 0.982 for the 2012 series and 0.975 for the 2021 series (class-specific accuracy, network architecture, and training procedures are provided in the Supporting Information; Table S2). However, cross-validation splits were made at the pixel rather than polygon level, and the same labelled polygons were applied to both years. These estimates therefore disproportionately reflect

accuracy near the training polygons; classification accuracy at unseen locations farther from the training data may be lower.

- **Georectification.** Ground control points were automatically extracted from orthorectified aerial photographs and a DEM to estimate camera parameters, and images were projected onto the 1 m analysis grid using the Python package `alproj` (<https://github.com/0kam/alproj>) (Okamoto et al., 2024). Median ground-control-point residuals were approximately 2 pixels, and the raster maps were completed with a single gap-filling pass using a 3×3 modal filter.

2.4. Quantification of *Sasa* change

We overlaid the 2012 and 2021 vegetation maps (Figure S5) and calculated a transition matrix for all classes. Throughout, we distinguish three quantities: **gross expansion**, the area transitioning from non-*Sasa* to *Sasa*; **gross loss**, the area transitioning from *Sasa* to non-*Sasa*; and **net change**, their difference. Areas are reported as the sum of the actual areas of 1 m cells measured on the Earth’s ellipsoid (geodesic area). These are nearly identical to nominal areas calculated as cell count $\times 1 \text{ m}^2$ (difference $< 0.1\%$); figure legends and model sample sizes refer to cell counts. Because *Sasa* can persist beneath shrub canopies, apparent loss may reflect canopy cover or shifts in the classification of mixed *Sasa*–shrub pixels rather than true decline. We report the destination classes of loss in the Results and address this ambiguity in the Discussion. To test whether apparent loss to dwarf pine was concentrated at boundaries, we calculated the Euclidean distance from every *Sasa*→dwarf pine cell to the nearest 2012 dwarf pine cell and binned the distances (1.5, 1.5–3, 3–5, and > 5 m), using all 2012 *Sasa* cells as the reference distribution; transition cells more than 5 m from 2012 dwarf pine were grouped into clusters by 8-neighbour connectivity and inspected individually (Table S4, Figure S6). To assess the nature of the transitions empirically, we conducted a blind interpretation of paired images. A total of 129 samples comprised 79 stratified samples of loss cells, proportional to the composition of destination classes, 40 stable controls (20 *Sasa* in both years and 20 dwarf pine in both years), and 10 reverse transitions (dwarf pine→*Sasa*). Paired image crops from 2012 and 2021 were presented in random order with transition types concealed; the interpreter was not told that stable controls and reverse transitions were interleaved among the loss cells. Each sample included two pairs, one from the green-leaf season and one from the autumn-colour season; frames with similar image greenness were paired to reduce misinterpretation arising from differences in leaf colour between acquisition dates. The first author assigned each sample to one of four interpretation categories: *Sasa* loss, canopy cover, boundary ambiguity, or indeterminate (full design and results in Table S5).

2.5. Snowmelt regime and trends

For every 1 m cell, we compiled 10 years of annual snowmelt DOY (2011–2021, excluding the missing year 2019) and calculated (i) **snowmelt climatology**,

the 2011–2021 mean snowmelt date (DOY), which averages over interannual variation to represent the prevailing snowmelt timing at each location, and (ii) a per-pixel linear trend, the slope of an ordinary least-squares regression of DOY against year, estimated for cells with at least 8 valid years. Per-pixel slopes were tested with two-sided t tests, and p values were adjusted across cells by the Benjamini–Hochberg procedure; at the landscape scale, we regressed the 10 annual spatial means of snowmelt DOY on year. As a perennial plant, *Sasa* is expected to respond to the prevailing local snowmelt regime rather than snowmelt in any single year, so both models used snowmelt climatology as the environmental predictor. Trends were not included as predictors. Per-pixel trend estimates are noisy, with a detection limit of approximately 2 days/year as discussed below, and adding the trend to Model B did not improve prediction in held-out spatial blocks (Table S1, Figure S4).

The trends require careful interpretation. With a 10-year record and a residual annual SD of approximately 9 days, the minimum detectable per-pixel slope is approximately 2 days/year; trends of plausible magnitude therefore cannot be detected statistically. Accordingly, trends were used only to parameterise scenarios for the 2030 projections (Section 2.7), not as evidence that a change in snowmelt had been demonstrated. Forecasting snowmelt timing itself was beyond the scope of this study.

2.6. Statistical models

We addressed the two questions set out in the Introduction, where environmental conditions permit *Sasa* and where observed establishment occurred, using two models with a shared evaluation design.

Shared predictors. Both models used slope, elevation, topographic position index (TPI; the difference between the elevation of a cell and the mean elevation of its eight neighbours, distinguishing ridges from hollows), topographic wetness index (TWI; $\ln(a/\tan \theta)$, with contributing area a and slope θ (Beven and Kirkby, 1979)), northness ($\cos$ aspect), and eastness ($\sin$ aspect), derived from the Fundamental Geospatial Data digital elevation model (DEM) of the Geospatial Information Authority of Japan (Geospatial Information Authority of Japan, 2026) (approximately 5 m mesh) and reprojected to the analysis grid, together with snowmelt climatology (Section 2.5). Slope, aspect, and TPI were computed with the GDAL `gdaldem` utility (slope and aspect by the Zevenbergen–Thorne algorithm (Zevenbergen and Thorne, 1987)); TWI was computed in QGIS with SAGA GIS (Conrad et al., 2015). The effective resolution of the terrain predictors is therefore that of the source DEM rather than 1 m; details of terrain-variable derivation are provided in the Supporting Information. Aspect, a circular variable, was decomposed into northness and eastness before reprojection. Terrain roughness and terrain ruggedness index were excluded before modelling because of strong collinearity with slope (Spearman $|r| = 0.93$ – 0.97). All retained predictors had variance inflation factors of at most 2.30 and pairwise $|r|$ below 0.7 (Table S7).

265 **Shared evaluation design.** A single set of spatial blocks was defined across
 266 the full analysis domain using `spatialsample` (Mahoney et al., 2023): a 10
 267 $\times$ 10 grid of rectangular blocks (each approximately 180×175 m) over the
 268 analysis extent, assigned at random to 4 folds, and used for tuning, ensemble
 269 construction, and held-out evaluation of both models. Model comparisons thus
 270 used the same evaluation areas.

271 2.6.1. Model A: environmental potential

272 Model A was a habitat suitability model fitted to the 2021 *Sasa* distribution.
 273 Presences were *Sasa* cells at 1 m resolution; pseudo-absences were non-*Sasa*
 274 cells thinned to one per 5 m cell to reduce class imbalance. We tuned four
 275 algorithms in `tidysdm` (Leonardi et al., 2024) by grid search with up to 18
 276 candidates per algorithm: gradient boosted trees (Friedman, 2001), MaxEnt
 277 (Phillips et al., 2006), random forest (Breiman, 2001), and generalised additive
 278 models. Candidates were selected by TSS on spatial blocks and blended using
 279 LASSO-regularised stacking (Couch and Kuhn, 2024). The final ensemble
 280 composition is reported in the Results. Performance is reported as the True Skill
 281 Statistic (TSS; Allouche et al. (2006)) and AUC on held-out spatial blocks. The
 282 main outputs were a continuous map of habitat suitability on a 0–1 scale (the
 283 suitability surface) and partial dependence profiles for each predictor, computed
 284 with the `DALEX` and `DALEXtra` packages. Where binarisation was needed, we used
 285 the threshold maximising TSS on held-out predictions; sensitivity of suitable
 286 area to thresholds of 0.1–0.7 is reported in Table S8. The main model used
 287 the full elevation range of the mapped slope. Results for a sensitivity variant
 288 restricting the domain to elevations below 2,560 m, the upper limit of the
 289 observed distribution, are provided in Table S6.

290 2.6.2. Model B: establishment model

291 Model B directly modelled observed transitions. For all cells that were non-*Sasa*
 292 in 2012 ($n = 1,197,536$; this count includes 5,147 cells that were unclassified
 293 in 2012, none of which became *Sasa*, whereas Table S3 is restricted to cells
 294 classified in both years), the response was whether they had become *Sasa* by
 295 2021 (establishment; $n = 4,095$). Predictors comprised the shared environmental
 296 variables and distance from the 2012 *Sasa* front. Distance was calculated as the
 297 Euclidean distance to the nearest cell belonging to a 2012 *Sasa* patch exceeding
 298 5 m^2 in geodesic area, with patches defined by 4-neighbour connectivity. This
 299 patch-size rule prevented tiny patches arising from classification noise from
 300 dominating the estimated expansion process: because area gain through rhizome
 301 elongation is proportional to patch perimeter, tiny patches with the highest
 302 perimeter-to-area ratios have disproportionate influence.

303 We fitted a binomial generalised additive model (GAM; `bam` in `mgcv` (Wood,
 304 2011)) to all eligible cells, using thin-plate spline smooths for all predictors. The
 305 distance term used a logarithmic transformation, `s(log(1 + d))`, concentrating
 306 smooth flexibility near the front, where observations were densest (criteria for
 307 specification choice are provided in Table S9). Basis dimensions were $k = 5$ for

the distance smooth and $k = 10$ for the environmental smooths. Model B was evaluated by leave-one-block-out refitting on the shared spatial partition. As a robustness comparison, we also fitted a gradient boosted tree ensemble to the same response and predictors on the same folds, using all establishment cells and a fold-stratified random sample of 150,000 non-establishment cells; because its negative set differs, its AUC is an approximate comparison. The estimated dispersal kernel is presented as mean predicted establishment probability alongside observed establishment rates in distance bands of 0–5, 5–10, 10–20, 20–40, 40–80, 80–160, and 160 m and beyond.

2.7. Projections to 2030

Model A’s suitability map describes the extent of environmentally permissive habitat, whereas rhizome expansion can proceed only from existing fronts. Applying the suitability map directly to the future would therefore greatly overestimate near-term establishment area. We instead projected establishment using a stochastic cellular automaton that applied Model B annually, and used Model A as the environmental-potential stratum corresponding to seed dispersal (Section 2.8). Projections thus arose from the expansion process itself rather than from a suitability threshold.

Starting from the observed 2021 distribution, we repeated the following steps for each year from 2022 to 2030. (i) Recalculate the distance surface from the current simulated distribution using the same 5 m² patch rule as in model fitting. (ii) Evaluate the 9-year establishment probability p_9 from Model B for every unoccupied cell and convert it to an annual probability: $p_1 = 1 - (1 - p_9)^{1/9}$. (iii) Draw establishment independently for each cell from a Bernoulli distribution, retaining occupancy once established (assuming no loss within the 9-year period). For computational efficiency, the environmental component of the linear predictor was precomputed for each scenario $\times$ year, and only the distance term was updated within the simulation; this decomposition is exact (Supporting Information). We ran 200 replicates per scenario (50 replicates for the spatial validation and structural sensitivity runs described below) and report per-cell establishment probabilities (means across replicates) and expected new establishment area (the sum of probability $\times$ cell area). The latter requires no binarisation threshold.

Scenarios. Snowmelt climatology was shifted linearly by $s \times (\text{year} - 2016)$, where 2016 is the midpoint of the averaging period. In the main text, we report $s = 0$ (no change, consistent with the absence of a detectable trend) and $s = -0.71$ days/year (the mean of the per-pixel slopes, close to the landscape-scale point estimate of -0.69 days/year); $s = -2.24$ days/year (the lower confidence limit of the landscape-scale trend) is reported as an extreme scenario in Table S11. These are scenarios conditional on the observed decade, not forecasts of snowmelt. Shifts were spatially uniform because per-pixel trend estimates are dominated by noise and cannot be directly extrapolated (Section 2.5). The uniform shift is an approximation, justified in the Supporting Information alongside an assessment of the spatial structure of per-pixel trends (Figure S4).

352 **Validation.** We conducted three checks. (1) *Hindcast*: we tested whether sim-
 353 ulations initialised from the 2012 distribution reproduced the observed 2012–2021
 354 gross expansion. Because Model B was fitted to the same interval, this is a
 355 calibration check rather than independent validation. (2) *Independent spatial*
 356 *validation*: for each spatial block, we refitted Model B excluding that block and
 357 scored the hindcast only within the omitted block. (3) *Structural sensitivity*:
 358 we quantified changes in the projections when the 5 m² patch rule was removed
 359 from distance updates and when annualisation was replaced by the first-order
 360 approximation $p_9/9$.

361 2.8. Risk to alpine communities

362 We present two risk strata corresponding to the two modes of *Sasa* expansion.
 363 The **near-term rhizome stratum** comprised cellular-automaton establishment
 364 probabilities restricted to cells classified as other vegetation in 2021, a class
 365 dominated by alpine herbs and including snowbed communities of conservation
 366 concern, summarised as expected establishment area. To identify the communities
 367 at risk, we overlaid the 1:25,000 vegetation GIS map compiled under the National
 368 Survey on the Natural Environment by the Biodiversity Center of Japan, Nature
 369 Conservation Bureau, Ministry of the Environment (Hokuriku block) ([Ministry](#)
 370 [of the Environment, 2026](#)) and cross-tabulated expected establishment area by
 371 mapped community. The scale mismatch between 1:25,000 polygons and the
 372 1 m grid is addressed as a limitation in the Discussion. The **seed-dispersal**
 373 **stratum** represented environmental potential in 2021, namely cells at or above
 374 the TSS-maximising threshold of Model A (Figure S7). Because establishment
 375 from seed is not constrained by the modelled distance process, environmental
 376 potential provides a first approximation of the area that could be exposed;
 377 unmeasured factors such as soils and constraints on seedling establishment are
 378 not accounted for. Field validation of community composition and assessment of
 379 rare-species occurrence within risk areas remain future work.

380 2.9. Software and reproducibility

381 All analyses used R 4.5.2 (terra, sf, tidymodels/tidysdm, stacks, spatialsample,
 382 mgcv, xgboost, ranger, maxnet, DALEX) in a scripted, non-interactive pipeline
 383 with fixed random seeds defined in a single configuration script; package versions
 384 are recorded in the `sessionInfo()` logs archived with the code. The vegetation
 385 classifier was implemented in Python with PyTorch, and image alignment used
 386 OpenCV. Code and data are archived at Zenodo ([https://doi.org/10.5281/zeno](https://doi.org/10.5281/zenodo.22745234)
 387 [do.22745234](https://doi.org/10.5281/zenodo.22745234)).

388 3. Results

389 3.1. *Sasa* expanded by a net +19% over 9 years

390 The area classified as *Sasa* (geodesic area) increased from 8,542 m² in 2012
 391 to 10,170 m² in 2021, a net change of +1,628 m² (+19%) over 9 years. This

net change reflects the difference between gross expansion of 4,095 cells (non-*Sasa*→*Sasa*; 48% of the 2012 area) and gross loss of 2,468 cells (*Sasa*→non-*Sasa*; 29%) (Figure 2; the full transition matrix is in Table S3; the nominal area of a cell is 1 m² and differs from its geodesic area by <0.1%). Expansion originated mainly from other vegetation (53%) and dwarf pine (35%), with the three deciduous shrub classes (rowans, maple, and montane alder) together accounting for 12%. Loss was predominantly to dwarf pine (47%) and other vegetation (38%), with the three deciduous shrub classes together accounting for 15%. Dwarf pine and *Sasa* exchanged area in both directions (interpretation in Section 4.1). In the blind interpretation of paired images, 73–87% of loss cells were assigned to boundary ambiguity, a proportion essentially unchanged from that in stable controls (80–85%). Canopy cover was assigned to 22% (8/37) of *Sasa*→dwarf pine transition cells, only slightly above the corresponding judgement for stable dwarf pine controls (15%, 3/20). Thus, the nature of transitions in individual 1 m cells could not be resolved at the photographic scale (Table S5). A few mapped 2021 *Sasa* cells were isolated from all 2012 *Sasa* cells, but these could not be distinguished from classification noise, and no isolated new patches were visually confirmed.

3.2. Snowmelt timing followed microtopographic gradients, with no detectable trend over 10 years

Snowmelt climatology ranged from approximately DOY 120 to 230 across the slope, following a ridge-to-hollow gradient (Figure S2). Per-pixel linear trends were negative in 91.7% of cells, positive in 8.3%, and zero in 0.06% (mean −0.71 days/year, median −0.63, SD 0.63; Figure S3), but annual spatial means varied markedly among years and provided little evidence for a landscape-scale trend (−0.69 days/year, 95% confidence interval −2.24 to +0.87, $p = 0.34$; Figure S1). Only 2.0% of cells had individually significant slopes ($p < 0.05$), and none remained significant after false-discovery-rate correction. Given the power limitations described in Section 2.5, the record for this period is consistent with a gradual advance in snowmelt but does not demonstrate it; trends are used below only to define scenarios.

3.3. Environmentally suitable area greatly exceeded occupied area (Model A)

The main Model A, covering the full elevation range, achieved a held-out TSS of 0.78 (mean across 4 spatial blocks; range 0.72–0.85) and an AUC of 0.95. Four members remained in the LASSO blend: 2 gradient boosted tree models, 1 MaxEnt model, and 1 random forest, with boosted trees and MaxEnt accounting for approximately 97% of the blend weight. Response curves (Figure 3) showed high habitat suitability at a snowmelt climatology of approximately DOY 150–185, peaking near DOY 172 and declining in later-melting snowbeds. Habitat suitability also declined with elevation and peaked at intermediate slopes. Responses to TPI, TWI, and aspect components were weaker and non-monotonic. Because estimates of suitable area depend strongly on the binarisation threshold, we primarily used the threshold of 0.16 that maximised

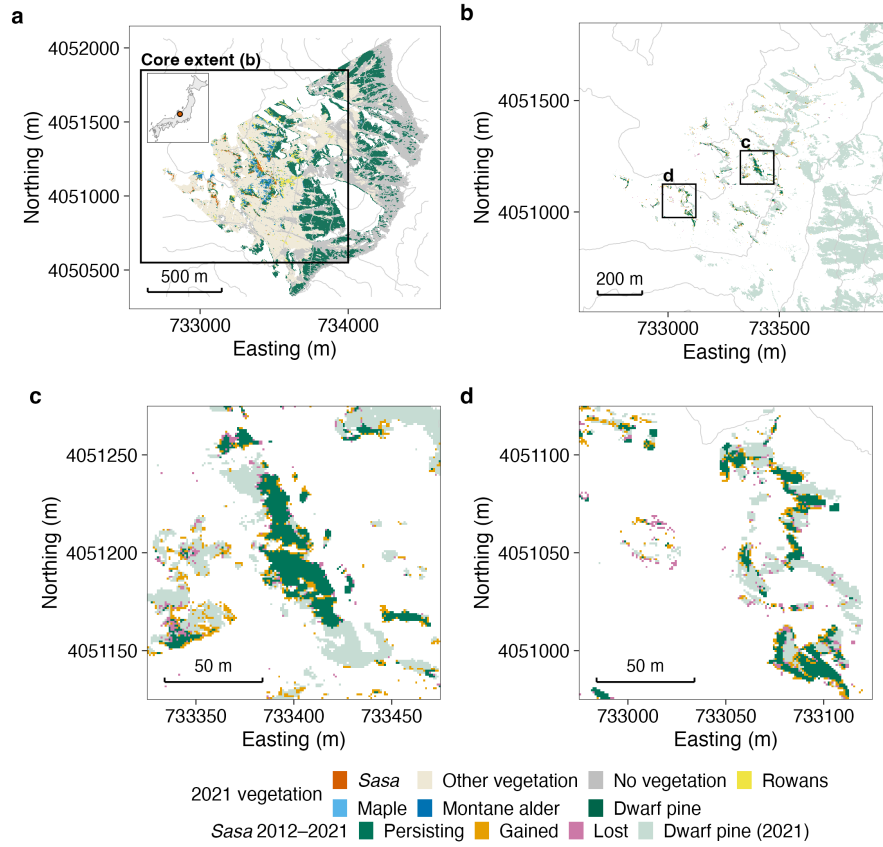


Figure 2: Changes in *Sasa* over 9 years were concentrated along the edges of existing patches. (a) Study site and 2021 vegetation map. The black box marks the core area in (b), and the inset locates the site within Japan. (b) *Sasa* change in the core area: persistence (*Sasa* in both years, 6,079 cells), gain (4,095 cells), and loss (2,468 cells). Pale green denotes dwarf pine in 2021, and white denotes all other classes. (c) A representative expansion front. (d) A representative *Sasa*→dwarf pine boundary; boxes in (b) locate the examples. Grey lines are 100 m contours. Area values are nominal areas based on cell counts.

435 TSS on held-out predictions (full threshold sensitivity in Table S8). At this
 436 threshold, suitable area was $164 \times 10^3 \text{ m}^2$, approximately 16 times the observed
 437 occupied area of $10.2 \times 10^3 \text{ m}^2$. Suitable area consistently exceeded occupied area
 438 at thresholds up to 0.55 (Figure 4 a).

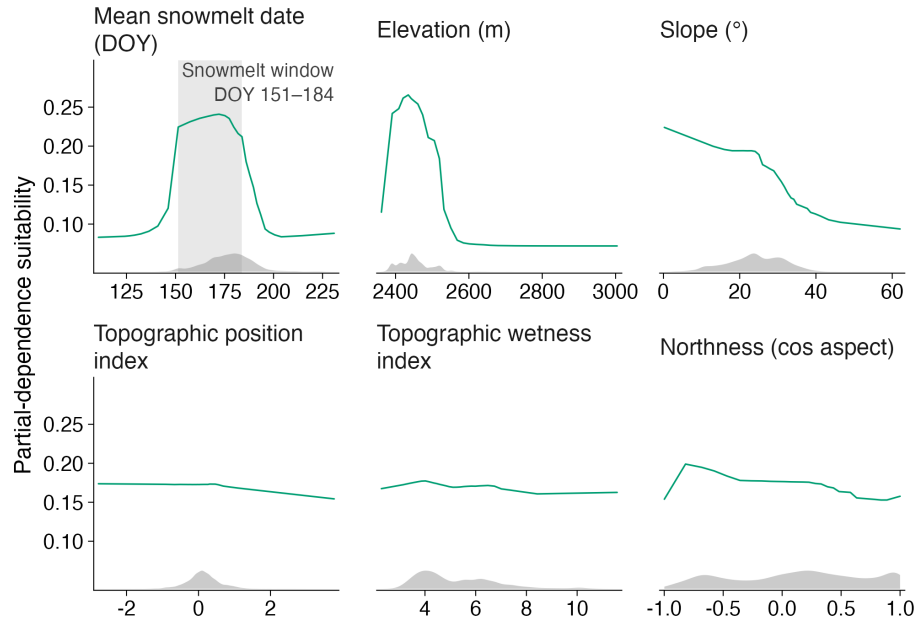


Figure 3: Model A habitat suitability was highest at a snowmelt climatology of approximately DOY 150–185, indicating a favourable snowmelt window for *Sasa* (shading; DOY 151–184, where suitability was at or above 85% of its peak). Each panel shows a partial dependence response curve for the main Model A variant. Grey bands along the bottom of the panels show the distribution (density) of each predictor in presence cells.

439 3.4. Establishment depended strongly on distance from the existing front (Model 440 B)

441 Of the 1,197,536 cells that were non-*Sasa* in 2012, 4,095 had become *Sasa* by
 442 2021. The establishment GAM discriminated well between establishment and
 443 non-establishment cells in leave-one-block-out evaluation (pooled AUC 0.946,
 444 TSS 0.75; block-specific AUC 0.91–0.96). A gradient boosted tree ensemble fitted
 445 on the same folds to all establishment cells and a fold-stratified sample of 150,000
 446 non-establishment cells achieved an AUC of 0.948, indicating that the additive
 447 model captured nearly all the signal available to a more flexible learner. Distance
 448 from the 2012 front was the strongest term. The 9-year establishment probability
 449 declined from approximately 10% within 5 m of the front to approximately 1% at
 450 5–10 m and below 0.1% beyond 80 m. Fitted distance-band probabilities closely
 451 matched observed establishment rates throughout the range (Figure 4 e; the
 452 largest discrepancy was 0.5 percentage points in the 0–5 m band). Environmental

associations were consistent with Model A: establishment probability was higher at late-melting (but not the latest-melting) sites, at lower elevations, and on gentler slopes. The map of 9-year establishment probability evaluated from the 2021 front (Figure 4 b) was concentrated in narrow bands around existing patches, with the highest values along their lower, late-melting edges.

3.5. Projected establishment continued near existing fronts, with limited sensitivity to snowmelt scenarios

The projection model reproduced the observed change before it was extrapolated. The cellular automaton initialised from the 2012 distribution reproduced the observed 2012–2021 gross expansion (expected 4,414 m², observed 4,093 m²; ratio 1.08; cell-level AUC 0.92). This was a calibration check. In independent spatial validation, in which each block was predicted by a model never exposed to that block, AUC ranged from 0.79 to 0.86, whereas block-level ratios of expected to observed area ranged from 0.67 to 2.0. Discrimination of establishment *location* was stable across blocks, but prediction errors in the *amount* of establishment were substantial.

Starting from the 2021 distribution, expected new establishment area by 2030 was 4,901 m² under the no-change scenario ($s = 0$; SD across replicates 70) and 5,730 m² under continued snowmelt advance at the observed rate ($s = -0.71$ days/year; SD 84), the latter 17% greater than the former (Figure 5). Both trajectories were nearly linear over 9 years, and both were of similar magnitude to the observed 2012–2021 gross expansion (geodesic area 4,093 m²). Establishment probability was highest along the lower, snowbed-facing edges of existing patches and negligible more than a few tens of metres from any patch (Figure 5 a, b). The extreme scenario at the lower confidence limit of the trend ($s = -2.24$ days/year) yielded 5,524 m², slightly *less* than the point-estimate scenario (Table S11). Projected area was not a monotonic function of earlier snowmelt (mechanism discussed in Section 4.2). Structural sensitivity of the projections is reported in Table S10.

3.6. Near-term risk was concentrated in two snowbed communities adjoining existing patches

Within cells classified as other vegetation in 2021, the expected establishment area by 2030 was 2,429 m² ($s = 0$) and 3,383 m² ($s = -0.71$ days/year) (Figure 6 a). Overlain on the Ministry of the Environment’s 1:25,000 vegetation map, 73–74% of this expected area fell in the *Fauria crista-galli*–*Carex blepharicarpa* snowbed community and a further 15–16% in the *Anaphalis alpicola*–*Phyllodoce aleutica* snowbed community; together, approximately nine tenths of the expected area lay in snowbed vegetation. The remaining tenth was mapped as dwarf pine community (Figure 6 b; Table S12). Observed 2012–2021 establishment had a similar composition (67%, 11%, and 22%, respectively; Table S12). The seed-dispersal stratum, Model A’s suitable area within other vegetation in 2021 at the TSS-maximising threshold, was more than an order of magnitude larger (Figure S7).

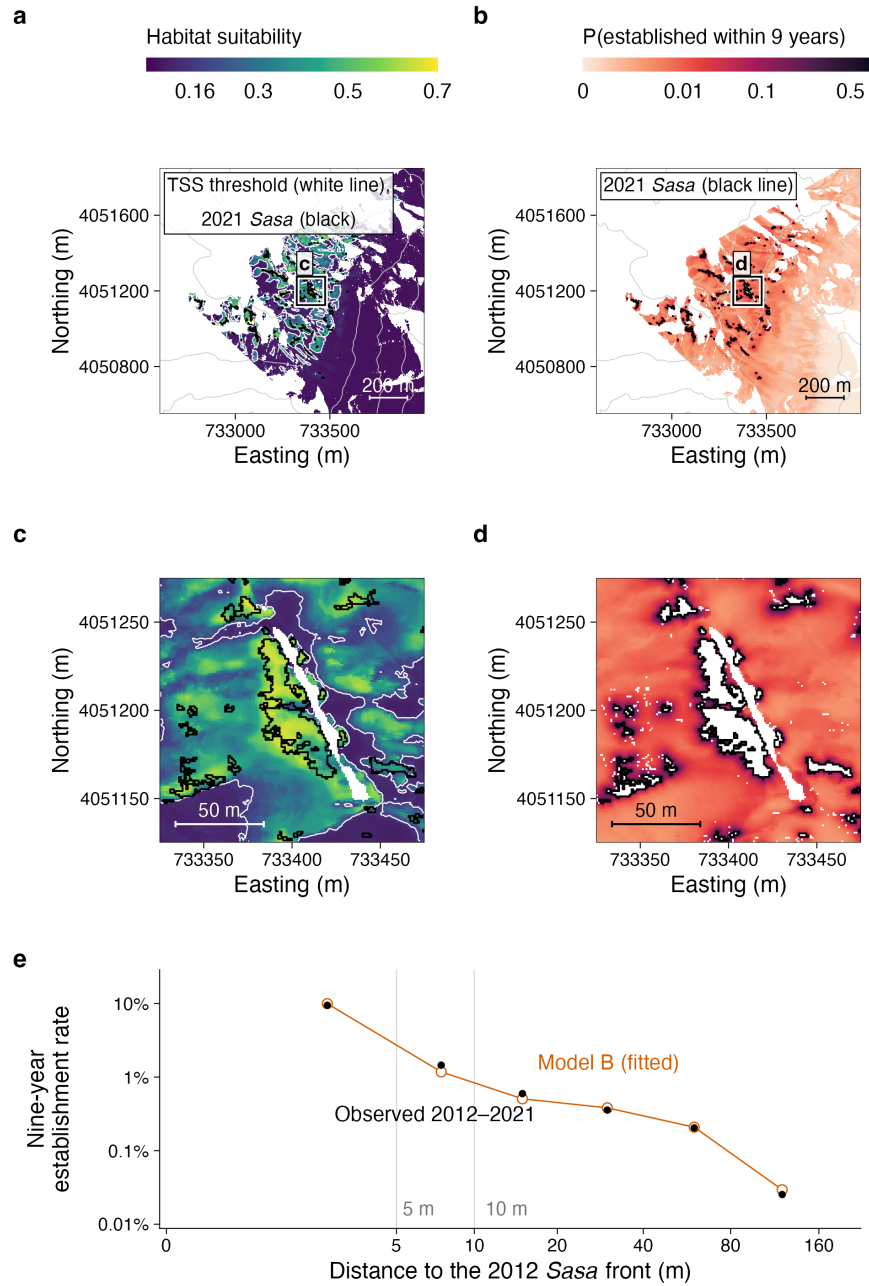


Figure 4: Environmentally suitable area greatly exceeded occupied area, but 9-year establishment was confined to a band within a few metres of the front. (a) Habitat suitability surface from the main Model A variant. White lines mark the TSS-maximising threshold (0.16), and black lines show the 2021 *Sasa* distribution, restricted to patches exceeding 5 m² in geodesic area, as in subsequent figures. (b) Model B 9-year establishment probability evaluated from the 2021 front. (c, d) Enlargements of the 150 m windows (black boxes) in (a) and (b), respectively. In (d), high establishment probabilities trace the edges of existing patches in bands one to a few cells wide. (e) Observed 9-year establishment rates (black points) and Model B fitted probabilities (orange; distance-band means connected by lines) by distance from the 2012 front. Both axes are logarithmic. The open-ended band at 160 m and beyond is omitted because it has no defined midpoint (observed establishment rate 0.0069%).

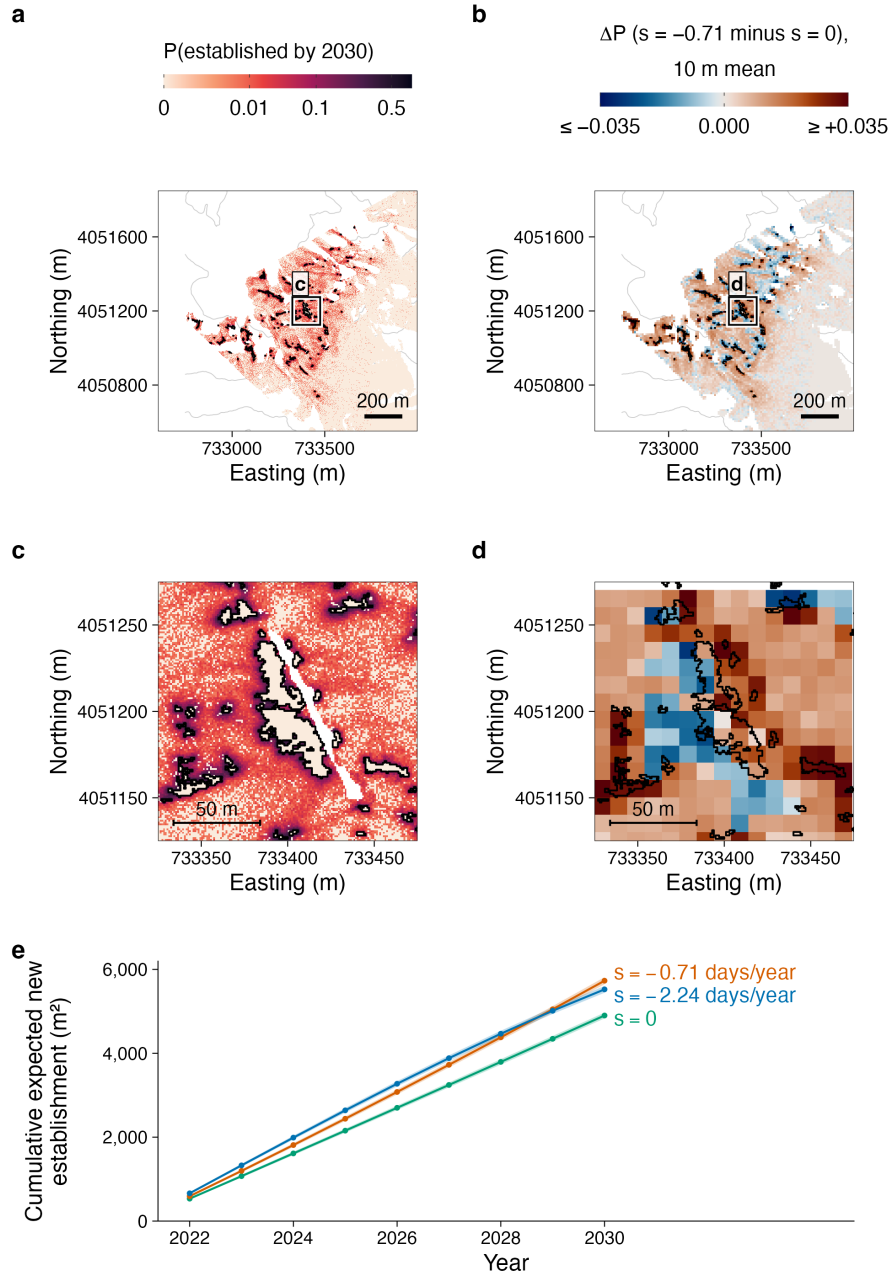


Figure 5: Projected establishment was confined to the vicinity of existing fronts, and the effect of the earlier-snowmelt scenario was small and spatially uneven. Projections to 2030 from the stochastic cellular automaton (200 replicates). (a) Establishment probability by 2030 under the no-change scenario ($s = 0$). (b) Difference between scenarios ($s = -0.71$ days/year minus $s = 0$, averaged over 10 m). Positive values (brown) indicate locations with higher establishment probability under earlier snowmelt; negative values (blue) indicate lower probability. (c, d) Enlargements of the 150 m windows (black boxes) in (a) and (b), respectively; (c) is at 1 m resolution and (d) shows 10 m averages. (e) Annual cumulative expected new establishment area for the 3 scenarios; ribbons show ± 1 SD across replicates. The $s = -2.24$ days/year trajectory remains above the point-estimate scenario through 2028 before crossing it in 2029. Black outlines show the 2021 *Sasa* distribution, restricted to patches exceeding $5 m^2$.

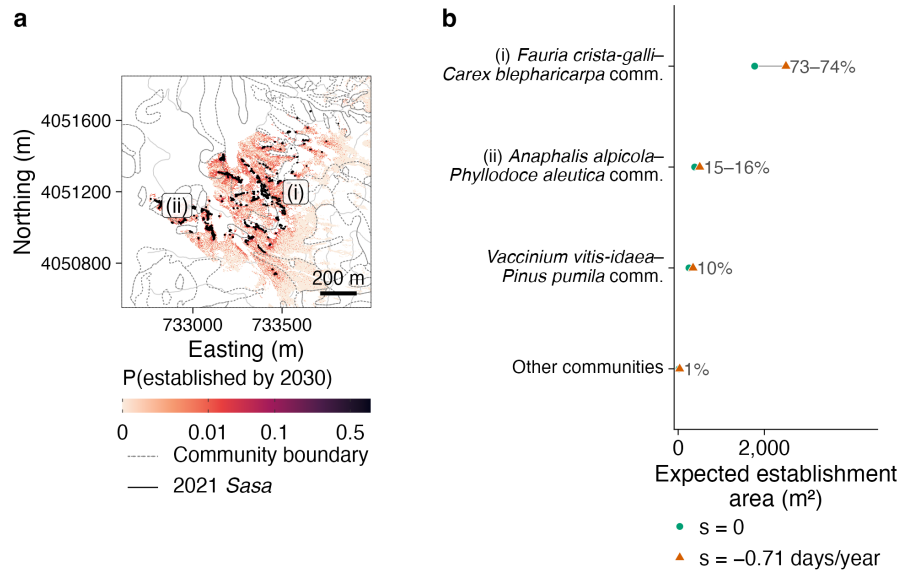


Figure 6: Near-term risk within herbaceous vegetation was concentrated in two snowbed communities adjoining existing patches. (a) Establishment probability by 2030 ($s = 0$) within cells classified as other vegetation in 2021. Dashed lines show community boundaries from the Ministry of the Environment's 1:25,000 vegetation map; (i) and (ii) locate the main communities shown in (b). (b) Expected establishment area by community (filled circles: $s = 0$; filled triangles: $s = -0.71 \text{ days/year}$). Percentage labels indicate each community's share of expected establishment area.

496 4. Discussion

497 Ground-based time-lapse imagery allowed us to map 9-year changes in *Sasa* on
 498 an alpine slope at 1 m resolution and to fit models and project expansion at the
 499 scale at which it occurs. Three findings emerge. First, *Sasa* expanded, but more
 500 modestly than the gross expansion figure alone suggests: net change was +19%
 501 over 9 years. Nearly half of the apparent loss was exchange with dwarf pine,
 502 and the interpretation of this exchange remains unresolved. Second, although
 503 snowmelt regime was among the strongest environmental predictors in both
 504 models, establishment probability was most strongly associated with proximity
 505 to existing patches. Environmentally suitable area far exceeded occupied area,
 506 suggesting that dispersal limitation, rather than a shortage of suitable habitat,
 507 accounts for the gap. Third, process-based projections calibrated to the observed
 508 9-year change indicated approximately 5,000–6,000 m² of spatially restricted
 509 establishment by 2030, concentrated in snowbed communities adjoining current
 510 patches and only modestly sensitive to earlier snowmelt. We discuss these findings
 511 in turn, followed by the role of the method, its limitations, and conservation
 512 implications.

513 4.1. Magnitude and nature of the observed change

514 The distinction between gross expansion, gross loss, and net change is sub-
 515 stantive. Gross expansion of 48% and net change of +19% describe the same
 516 9 years; the difference comprises 2,468 cells classified as *Sasa* in 2012 but not
 517 in 2021. Of this loss, 47% was to dwarf pine and 38% to other vegetation; the
 518 three deciduous shrub classes (rowans, maple, and montane alder) accounted
 519 for only 15%. Transitions to dwarf pine require careful interpretation. Dwarf
 520 pine is a prostrate, slow-growing conifer with annual shoot extension of only
 521 approximately 4–5 cm (Amagai et al., 2015). Lateral advance over 9 years would
 522 amount to at most approximately 0.5 m, less than a single 1 m cell, making
 523 extensive concealment of *Sasa* stands by dwarf pine unlikely. Dwarf pine stands
 524 on the study slope are themselves expanding (Amagai et al., 2015), however,
 525 and their edges interdigitate with *Sasa* at sub-cell scales. Two mechanisms are
 526 therefore consistent with the data: lateral growth of dwarf pine branches into
 527 boundary cells and shifts in the classification of mixed pixels near boundaries.
 528 Indeed, 87.5% of *Sasa*→dwarf pine transition cells directly adjoined dwarf pine
 529 present in 2012 (median distance 1.0 m), and isolated transitions at least 5 m
 530 from the front accounted for only 1.6% of transition area (19 m²; Table S4,
 531 Figure S6). Thus, almost all apparent loss to dwarf pine occurred at dwarf
 532 pine boundaries, although the blind interpretation of paired images could not
 533 distinguish lateral growth from shifts in classification. Judgement distributions
 534 for loss cells were almost identical to those for stable controls (Section 3.1), and
 535 the fate of individual 1 m cells could not be resolved at the photographic scale.
 536 The boundary interpretation therefore rests on the concentration of adjacent
 537 transitions and the constraint imposed by dwarf pine growth rates. Accordingly,
 538 we treat loss area as an upper bound on true decline and net change as a conser-

vative estimate of expansion. In contrast, transitions to other vegetation may include genuine local retreat and warrant targeted field verification.

The observed rate, +19% across the slope over 9 years (approximately 2%/year), lies at the lower end of plot-based estimates from Murodo-daira, where *Sasa* patches expanded by 44–260% over 38 years (Yoshida et al., 2016). In the Daisetsuzan Mountains, *Sasa kurilensis* cover in snowbed meadows reportedly increased by up to 47% over 32 years, with fronts advancing at a mean of 39 cm/year and a maximum of 50 cm/year (Kudo et al., 2011); on an annual basis, our observations are of similar magnitude. Continuous mapping across the slope reproduces the direction and order of magnitude reported by plot studies while adding spatial structure that plots cannot provide: where fronts move and where they remain stationary.

4.2. Roles of dispersal limitation, environment, and snowmelt

Distance from the existing front dominated the establishment model: establishment probability fell by an order of magnitude within the first 10 m of the front (Section 3.4). This distance scale is consistent with the rates of front advance reported for *Sasa* in the Daisetsuzan Mountains (Section 4.1; Kudo et al. (2011)). The strong distance dependence offers a direct explanation for why environmental potential (Model A) exceeded occupied area across a broad range of thresholds, reaching approximately 16 times the occupied area at the TSS-maximising threshold. Taken together, this gap between suitable and occupied habitat and the slow advance of fronts are consistent with a non-equilibrium interpretation in which the current *Sasa* distribution has yet to catch up with its environmental potential. If suitable habitat has expanded first, for example in association with earlier snowmelt, and dispersal-limited fronts are still filling that space, a stronger association with proximity than with environment would be expected.

Environmental terms describe variation in the rate of expansion from these fronts. In both models, habitat suitability and establishment probability peaked at a snowmelt climatology of approximately DOY 170–175 (Model A peak 172; Model B peak 175) and declined towards both earlier- and later-melting sites. *Sasa* thus has a favourable snowmelt window, and many current fronts adjoin sites on the later-melting side of that window. This response is consistent with the mechanism proposed for Daisetsuzan populations, in which a longer growing season accompanies *Sasa* advance from dwarf pine margins into wetter, later-melting sites (Kudo et al., 2011). It is also consistent with snow-manipulation experiments indicating that snowmelt timing structures tundra and alpine plant communities through growing-season length, soil temperature, and moisture (Wipf and Rixen, 2010; Wahren et al., 2005). Snowmelt DOY is nevertheless a proxy for these covarying conditions and was measured, not manipulated, in this study. Modelled associations are not evidence of causality; separating direct snowmelt effects from those of soils and microclimate requires field measurements.

581 The snowmelt-window interpretation also offers an explanation for the appar-
 582 ently unusual behaviour of the projections. Projected establishment was 17%
 583 greater under the point-estimate earlier-snowmelt scenario, but slightly lower
 584 under the extreme scenario. Sites immediately adjacent to the front, where
 585 establishment is concentrated, lie on the late-melting side of the window, so
 586 a moderate advance shifts them towards its centre, whereas a larger advance
 587 begins to move them beyond it. The response of expansion to snowmelt is
 588 therefore non-monotonic, and its sign depends on the position of the front along
 589 the snowmelt gradient. This result becomes apparent only when expansion is
 590 modelled as a process.

591 4.3. *Is Sasa expansion a form of shrubification?*

592 Woody shrub expansion is among the best-documented vegetation responses to
 593 warming in Arctic and alpine tundra (Myers-Smith et al., 2011; Bjorkman et al.,
 594 2018; Carlson et al., 2017). Although *Sasa* is not a woody shrub, it behaves
 595 functionally like expanding tundra shrubs: it forms dense, tall, evergreen stands
 596 that exclude subordinate species (Kudo et al., 2011), and snowbed vegetation
 597 recovers after *Sasa* removal (Kudo et al., 2017). It differs from typical shrubs in
 598 two respects, however. First, rhizome expansion proceeds from existing fronts
 599 with a metre-scale dispersal kernel, making it more spatially predictable than
 600 the spread of shrubs that establish new nuclei through seed dispersal and, in
 601 principle, amenable to targeted removal at the front. Second, *Sasa* competes
 602 with dwarf pine, which is also expanding under warming (Amagai et al., 2015).
 603 Because most of the bidirectional *Sasa*-dwarf pine exchanges in the transition
 604 matrix occurred at boundary cells whose nature we could not resolve, whether
 605 the two expanding taxa also replace each other remains an open question that
 606 merits study in its own right.

607 4.4. *Ground-based cameras among vegetation monitoring approaches*

608 The role of ground-based cameras in this framework is complementary to that
 609 of satellites. Satellite time series offer continuity over decades and comparability
 610 among regions, making them well suited to analyses of landscape-scale greening
 611 and range shifts. Ground-based cameras provide sub-metre resolution and daily
 612 revisits within a fixed field of view, matching the scale of processes underlying the
 613 changes observed by satellites: patch-front advance, the sequence of snowmelt
 614 within snowbeds, and shifts in community boundaries. Our study illustrates what
 615 becomes possible in this gap: an establishment model fitted at the resolution of
 616 the process. The limitations are also specific: a fixed oblique viewpoint, coverage
 617 of at most a few km², dependence on continuous imagery throughout the period
 618 of interest, and restriction to taxa distinguishable in photographs, such as large
 619 or phenologically distinctive plants, including *Sasa* and dwarf pine. The last
 620 limitation could be partly relaxed through field collection of training data.

621 4.5. *Limitations*

622 (1) *Two time points.* The establishment model was fitted to a single 9-year
 623 interval, and projections assume stationary annual establishment rates

conditional on the covariates. Independent validation was limited to spatial block exclusion, which indicated that *where* establishment occurs can be predicted reasonably well in unseen areas (AUC 0.79–0.86), whereas errors in *how much* occurs are substantial (block-level ratios 0.7–2.0). (2) *Snowmelt trends*. A 10-year record cannot detect trends of plausible magnitude. The scenarios evaluate no change, the point estimate of the observed slope, and its lower confidence limit; they are not predictions of snowmelt itself. (3) *Patch rule*. Projections depend strongly on the assumption that new colonies contribute to expansion only after forming an established patch ($>5 \text{ m}^2$); removing this rule triples projected area. The biological rationale is given in Section 2.6, but the threshold is not a measured parameter. (4) *Classification*. Map accuracy estimates are upper bounds (see Section 2.3), and the *Sasa*–dwarf pine exchanges show that boundary pixels are the most uncertain parts of the maps. These are precisely the pixels that define the front. (5) *Environmental coverage*. Soil properties, soil temperature, moisture, and freeze–thaw exposure were not measured. The strong distance term may partly absorb the effects of unmeasured, spatially structured variables. (6) *Seed dispersal*. The establishment model captures only rhizome expansion. Establishment from seed following flowering events (Miyazaki et al., 2009; Mizuki et al., 2014) bypasses the distance constraint, which is why environmental potential is presented as a separate risk stratum. Even when seeds arrive, however, germination and seedling establishment may impose constraints independent of dispersal.

4.6. Conservation implications

Two practical implications follow. First, near-term risk is spatially concentrated and can be assigned to specific locations. Approximately nine tenths of expected establishment within herbaceous vegetation by 2030 is projected to fall in two snowbed communities on the Ministry of the Environment’s vegetation map, the *Fauria crista-galli*–*Carex blepharicarpa* and *Anaphalis alpicola*–*Phyllodoce aleutica* communities, and to lie within a few metres of existing patches. The remaining tenth, allocated to the dwarf pine community, reflects herbaceous cells within coarse 1:25,000 polygons rather than establishment beneath dwarf pine canopies. Removal or containment at patch fronts bordering snowbeds is therefore a high-priority candidate intervention; intervention costs and regrowth following removal were not evaluated here. Second, the much larger environmentally suitable area, most of which is unreachable by rhizomes within the projection horizon, is where establishment from seed would matter. Targeted surveillance for new isolated patches after flowering events, rather than continuous monitoring of the entire slope, would be a proportionate response. The presence or absence of rare species within risk areas cannot be determined at the resolution of the Ministry’s 1:25,000 vegetation map; field surveys are the natural next step.

More generally, this framework separates two questions that a single suitability map can conflate: where environmental conditions permit a species, and how far

it can actually reach. It answers the latter with a process model calibrated to observed change. For species such as *Sasa* that expand at metre scales, fixed cameras make this calibration possible at the scale of the process itself. The study slope is a maritime, monsoon-influenced alpine environment with exceptionally deep snow; in continental settings with shallower snowpack and greater freeze–thaw exposure, the balance between snowmelt and other constraints may differ. The same logic could nevertheless apply to other expanding species in snowy environments where comparable image records are available.

5. Conclusion

Over 9 years, *Sasa* expanded on the study slope with a net change of +19% (gross expansion 48%, gross loss 29%). Establishment probability was most strongly associated with proximity to existing patches, with further variation associated with snowmelt regime and topography. Environmentally suitable area greatly exceeded occupied area, suggesting that dispersal limitation accounts for the gap. A process model calibrated to observed change projected approximately 4,900–5,700 m² of further establishment by 2030 under scenarios conditioned on the observed decade rather than forecasts of snowmelt, concentrated in snowbed communities adjoining current patches and only modestly sensitive to earlier snowmelt. Using *Sasa* as an example, this study shows that ground-based time-lapse imagery fills the gap between plot surveys and satellite observations, allowing metre-scale expansion to be observed and projected at the resolution of the process itself.

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Conflict of Interest Statement

The authors declare no conflicts of interest.

Author Contributions

Ryotaro Okamoto: Conceptualization (equal); Data curation (lead); Methodology (lead); Software (lead); Writing – original draft (lead); Writing – review & editing (equal). **Hiroyuki Oguma:** Conceptualization (equal); Investigation (equal); Project administration (lead); Supervision (lead); Writing – review & editing (equal). **Reiko Ide:** Conceptualization (equal); Data curation (equal); Writing – review & editing (equal).

701 Data Availability Statement

702 The imagery-derived vegetation and snowmelt rasters, terrain predictors, all
703 fitted model objects, projection outputs, and scripted analysis pipeline with
704 its acceptance tests are permanently archived at Zenodo (<https://doi.org/10.5281/zenodo.22745234>). A development mirror is maintained at <https://github.com/Okam/SasaSDMPaper>.
706

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